COHOMOLOGY OF LIE ALGEBRAS AND THE LEVI DECOMPOSITION

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Abstract. The goal of this talk is to formally introduce and prove the Levi decomposition of a Lie algebra over a field of characteristic zero. To this end, we construct the cohomology of a Lie algebra starting from the Lie group setting. Various auxillary results are established such as the Chevalley-Eilenberg complex and Whitehead's theorem, which is proved via an adaption of [Web13]. This work is heavily based on Etingof's Lie Groups and Lie Algebras lecture notes [Eti24], with major additions to the exposition and proof of the Levi decomposition itself.

Contents

1. The Chevalley-Eilenberg Complex of Lie Groups

The cohomology of a Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)_{\mathbb{C}}$ is given by the cohomology of the Chevalley-Eilenberg complex of $\mathfrak{g}$, which is first defined on Lie groups G over a manifold M. We can study these structures using the de Rham cohomology

$$H^i(M;\mathbb{C}) = \Omega^i_{\text{closed}}(M)/\Omega^i_{\text{exact}}(M)$$

derived from the chain complex of smooth differential i-forms on M

$$0 \longrightarrow \Omega^0(M) \xrightarrow{d_0} \Omega^1(M) \xrightarrow{d_1} \Omega^2(M) \xrightarrow{d_2} \cdots \xrightarrow{d_{n-1}} \Omega^n(M) \xrightarrow{d_n} 0$$

where $\Omega^i_{\text{closed}} = \ker(d_i)$ and $\Omega^i_{\text{exact}} = \mathrm{im}(d_{i-1})$. If M is endowed with a vector field v, the Lie derivative, given by

$$L_v \colon \Omega^*(M) \to \Omega^*(M)$$

is the *unique* derivation of the algebra of differential forms commuting with de Rham differentials. Geometrically, L_v measures the rate of change of a geometric object as we drag it along the flow of v. For differential forms, the most practical tool to use is Cartan's Magic Formula in which

$$L_v\omega = \iota_v(d\omega) + d(\iota_v\omega)$$

where ι_v is the interior product (or contraction operator). This guy has some nice properties and could be the subject of his own talk, but we must continue on. The averaging operator $P\colon \Omega^*(M) \to \Omega^*(M)$ is a tool used to create a form which is invariant under flows of v (or generally a group action):

$$P(\omega) = \int_G g^*\omega dg$$

where g^* is a pullback and dg the normalized Haar measure. So if G is compact acting on M, then P commutes with d and satisfies $P^2 = P$, admitting the decompoition

$$\Omega^*(M) = \mathrm{im}(P) \oplus \ker(P) = \Omega^*(M)^G \oplus \Omega^*(M)_0.$$

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Here, $\Omega^*(M)^G$ consists of the G-invariant forms ($g^*\omega = \omega$ for $g \in G$) and $\Omega^*(M)_0$ consists of the zero-average forms (cancelled out by symmetry of G).

$$\begin{tikzcd}
\cdots \arrow[r] & \Omega^{i-1}(M) \arrow[r, "d_{i-1}"] \arrow[d, "P"] & \Omega^i(M) \arrow[r, "d_i"] \arrow[d, "P"] & \Omega^{i+1}(M) \arrow[r, "d_{i+1}"] \arrow[d, "P"] & \cdots \\
\cdots \arrow[r] & \Omega^{i-1}(M)^G \arrow[r, "d_{i-1}"] & \Omega^i(M)^G \arrow[r, "d_i"] & \Omega^{i+1}(M)^G \arrow[r, "d_{i+1}"] & \cdots
\end{tikzcd}$$

It turns out that the complex $\Omega^*(M)_0$ is exact since P induces an isomorphism in cohomology. This leads to our first important result.

Theorem 1.1. The cohomology $H^*(M;\mathbb{C})$ is computed by the complex of invariant differential forms $\Omega^*(M)^G$. If G is a compact Lie group, then $H^*(G;\mathbb{C})$ is computed from the complex of $\Omega^*(G)^G$ of left-invariant differential forms on G, known as the Chevalley-Eilenberg complex of G.

This proof of this is not necessarily trivial and, so as not to divert from the main goal of this talk, is omitted for now.

2. Cohomology of Lie Algebras

Recall from elementary differential geometry the invariant formula for the exterior derivative given for $\omega \in \Omega^n(M)$ and vector fields $v_0, \ldots, v_n$ on M:

$$d\omega(v_0,\ldots,v_n) = \sum_i (-1)^i L_{v_i}(\omega(v_0,\ldots,\widehat{v_i},\ldots,v_n)) + \sum_{i<j}(-1)^{i+j}\omega([v_i,v_j],v_0,\ldots,\widehat{v_i},\ldots,\widehat{v_j},\ldots,v_n)$$

where the hat indicates omitted terms. In fact, since the functions $\omega(v_0\ldots,\widehat{v_i},\ldots,v_n)$ are constant when considering $\omega \in \Omega^n(G)^G$ for Lie group G, we can boil this down to

$$d\omega(v_0,\ldots,v_n) = \sum_{i<j}(-1)^{i+j}\omega([v_i,v_j],v_0,\ldots,\widehat{v_i},\ldots,\widehat{v_j},\ldots,v_n).$$

To define the Chevalley-Eilenberg complex of a Lie algebra $\mathfrak{g}$, we'll make use of the following fact.

Proposition 2.1. For Lie group G and $\mathfrak{g} = \mathrm{Lie}(G)_{\mathbb{C}}$, we have $\Omega^n(G)^G \cong \wedge^n\mathfrak{g}^*$.

Proof. Forms $\omega \in \Omega^n(G)^G$ are left-invariant, meaning

$$\omega_g(v_1,\ldots,v_n) = \omega_e\big((dL_{g^{-1}})_g v_1,\ldots,(dL_{g^{-1}})_g v_n\big)$$

for every $g \in G$, so the value of ω at any g is determined by its value at e. At the identity,

$$\omega_e \in \wedge^n T_e^*G = \wedge^n\mathfrak{g}^*$$

which induces a map $\Omega^n(G)^G \to \wedge^n\mathfrak{g}^*$ given by $\omega \mapsto \omega_e$. The reverse direction is given by taking $\alpha \in \wedge^n\mathfrak{g}^*$ and defining

$$\omega_g(v_1,\ldots,v_n) := \alpha\big((dL_{g^{-1}})_g v_1,\ldots,(dL_{g^{-1}})_g v_n\big)$$

which is smooth, alternating, and satisfies left-invariance. These maps are inverses, forming an isomorphism in the category of vector spaces. $\square$

Corollary 2.1. For a Lie group G and Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)_{\mathbb{C}}$, the complex of left-invariant forms $\Omega^*(G)^G$ is the same as the complex given by

$$0 \longrightarrow \mathbb{C} \longrightarrow \mathfrak{g}^* \longrightarrow (\wedge^2\mathfrak{g})^* \longrightarrow \cdots \longrightarrow (\wedge^n\mathfrak{g})^* \longrightarrow \cdots$$

with the differential defined above ($d^2 = 0$ following from the Jacobi identity) endowed with wedge product multiplication. This is the **Chevalley-Eilenberg Complex** of the Lie algebra $\mathfrak{g}$, which we denote $CE^*(\mathfrak{g})$.

The cohomology of $CE^*(\mathfrak{g})$ is the Lie algebra cohomology $H^*(\mathfrak{g})$. If we now take a $\mathfrak{g}$-module V to pick coefficients from, we define

$$CE^*(\mathfrak{g}; V) := \mathrm{Hom}(\wedge^* \mathfrak{g}, V)$$

with differential given by the exterior derivative above. This generates the cohomology $H^*(\mathfrak{g}; V)$. In fact if V is finite dimensional and $\mathfrak{g} = \mathrm{Lie}(\mathfrak{g})_{\mathbb{C}}$, then

$$CE^*(\mathfrak{g}; V) := (\Omega^*(G) \otimes V)^G$$

with de Rham differential. We arrive at an important auxillary result for the characterization of Lie algebra cohomology and the proof of the Levi decomposition.

Theorem 2.1 (Whitehead)**.** If $\mathfrak{g}$ is any semisimple Lie algebra and V a finite dimensional $\mathfrak{g}$-module, then we have that

$$H^1(\mathfrak{g}; V) = H^2(\mathfrak{g}; V) = 0.$$

Proof. We'll only prove the result for H^2 since H^1 was covered in class :) Given a basis $\{g_i\}$ of $\mathfrak{g}$ we can create the dual basis $\{g^i\}$ forcing the Killing form to satisfy

$$K_{\mathfrak{g}}(g_i, g^i) = \delta_{ij}$$

which is non-degenerate over semisimple Lie algebras. We construct the Casimir element $c\colon V \to V$ to be the operator

$$c = \sum_i g_i g^i$$

which commutes with the action of Lie algebra $\mathfrak{g}$ whence $[c, \mathfrak{g}] = 0$. Further, we can assume that V is irreducible since otherwise

$$V = \bigoplus_i V_i \quad \implies \quad H^*(\mathfrak{g}; V) = \bigoplus_i H^*(\mathfrak{g}; V_i).$$

By Schur's Lemma we know that $c \curvearrowright V_i$ by constant and by zero if $\mathfrak{g} \curvearrowright V$ trivially (that is, V consists of invariant elements). With this out of the way, take a 2-cocycle $\omega \in Z^2(\mathfrak{g}; V)$ such that

$$0 = d_2\omega = x \cdot \omega(y, z) + \curvearrowright + \curvearrowright - \big(\omega([x, y], z) + \curvearrowright + \curvearrowright\big).$$

Suppose first that $c \neq 0$. Then we can define the 1-cochain $h\colon \mathfrak{g} \to V$ given by

$$h(x) = c^{-1} \sum_i g_i \cdot \omega(g^i, x).$$

Setting $z = g^i$ and applying the action of g_i to $d_2\omega$, we get terms such as

$$\sum_i g_i \cdot \big(g_i \cdot \omega(x, y)\big) = c\omega(x, y)$$

and then some remaining terms such as

$$\sum_i g_i \cdot x \cdot \omega(y, g^i), \quad \sum_i g_i \cdot y \cdot \omega(g^i, x)$$

which are "module action" parts and terms such as

$$\sum_i g_i \cdot \omega([x, y], g^i), \quad \sum_i g_i \cdot \omega([y, g^i], x), \quad \sum_i g_i \cdot \omega([g_i, x], y)$$

which are "bracket" parts. Recall that a property of a dual basis under the Killing form $K_{\mathfrak{g}}$ is that if $[x, g_i] = \sum_i \alpha_i^j g_j$ and $[x, g^i] = \sum_i \beta_j^i g^j$ then $\alpha_i^j + \beta_j^i = 0$. Hence "error" terms where the action of $\mathfrak{g}$ hits basis elements cancel when summed, leaving the identity

$$c\omega(x, y) = \sum_i g_i \cdot \omega([x, y], g^i) - x \cdot \sum_i g_i \cdot \omega(y, g^i) - y \cdot \sum_i g_i \cdot \omega(g^i, x).$$

Now $cd_1h = c\omega$ and since $c \neq 0$ we have $d_1h = \omega$ making it a 1-coboundary. The other case is when $c = 0$, leading to $\mathfrak{g} \curvearrowright V$ trivially and hence $V \cong \mathbb{C}$ by complete reducibility. In this case, half the terms of $d_2\omega$ vanish, leaving us with

$$0 = d_2\omega = \omega([x, y], z) + \curvearrowright + \curvearrowright$$

as the 2-cocycle condition. Considering now $\mathfrak{g}^*$ as a $\mathfrak{g}$-module under dual representation such that, for any 2-form like ω, we automatically have a $\mathfrak{g}^*$-values 1-form $A \in \mathrm{Hom}(\mathfrak{g}, \mathbb{C})$ via

$$A(x)(l) = \omega(l, x).$$

Observe that

$$\begin{aligned} d_1A(x, y)(l) &= x \cdot (A(y))(l) - y \cdot (A(x))(l) - A([x, y])(l) \\ &= -A(y)([x, l]) + A(x)([y, l]) - A([x, y])(l) \\ &= -\omega([l, x], y) + \curvearrowright + \curvearrowright \end{aligned}$$

whence ω bein a closed form implies that A is also a closed form! By application of Whitehead's Theorem in the H^1 case, A must also be exact such that $A = d\rho$ for some $\rho \in Z^0$, and by back substitution

$$\omega(l, x) = A(x)(l) = \rho([l, x])$$

meaning ω is also exact as desired. □

We can extract rich meaning from $H^i(\mathfrak{g}; V)$ especially for $i = 0, 1, 2$, as we'll come to find. Heuristically, we're measuring how badly $\mathfrak{g}$ fails to be simple or interact with other structures by means of "obstructions" or "holes" in the algebraic structure. Classifying such $H^i(\mathfrak{g}; V)$ will turn out to be a fruitful (and fun) endevour.

Example 2.1 ($i = 0$)**.** We have that $H^0(\mathfrak{g}; V) = H(CE^0(\mathfrak{g}; V)) = H(\mathrm{Hom}(\wedge^0\mathfrak{g}, V)) = V^{\mathfrak{g}}$ which are the $\mathfrak{g}$-invariants in V.

Example 2.2 ($i = 1$)**.** A linear map (1-form) $\omega \in \mathrm{Hom}(\mathfrak{g}, V)$ is a 1-cocycle if $d_1\omega = 0$, defined as

$$d_1\omega(x, y) = x \cdot \omega(y) - y \cdot \omega(x) - \omega([x, y])$$

which vanishes for $\omega([x, y]) = x\omega(y) - y\omega(x)$. Intuitively, if $x\omega(y) - y\omega(x)$ is the difference in the value of ω when moving alone x then y, versus y then x as infinitesimal flow, then $\omega([x, y])$ is the correction term needed to close the gap of the quadrilateral. Similarly, a 0-form $v \in V$ forms a boundary of ω if $\omega(x) = xv$. Whence $H^1(\mathfrak{g}; V) = Z^1(\mathfrak{g}; V)/B^1(\mathfrak{g}; V)$ in terms of these maps.

Remark. $Z^1(\mathfrak{g}; \mathfrak{g})$ represents a Lie algebra of derivations of $\mathfrak{g}$, and $B^1(\mathfrak{g}; \mathfrak{g})$ the ideal of inner derivations! Hence $H^1(\mathfrak{g}; \mathfrak{g})$ is the remaining outer derivations of $\mathfrak{g}$. That is, all derivations of a *semisimple* Lie algebra $\mathfrak{g}$ are inner (that $H^1(\mathfrak{g}; \mathfrak{g}) = 0$).

Example 2.3 ($i = 2$)**.** A 2-form $\omega \in \mathrm{Hom}(\wedge^2\mathfrak{g}, V)$ is a 2-cocycle if $d_2\omega = 0$, which is when

$$d_2\omega(x, y, z) = x \cdot \omega(y, z) - y \cdot \omega(x, z) + z \cdot \omega(x, y) - \omega([x, y], z) + \omega([x, z], y) - \omega([y, z], x) = 0.$$

This is equivalent to requiring that the bracket

$$[(x, a), (y, b)] = \big([x, y], xb - ya + \omega(x, y)\big)$$

defines a Lie algbra structure on $\mathfrak{g} \oplus V$, which holds true by application of the Jacobi identity, yielding $d_2\omega(x, y, z) = 0$. Such is the case for any abelian extension $\widetilde{\mathfrak{g}}$ of $\mathfrak{g}$ by V such that

$$0 \longrightarrow V \longrightarrow \widetilde{\mathfrak{g}} \longrightarrow \mathfrak{g} \longrightarrow 0$$

for an abelian ideal V. Two extensions $\widetilde{\mathfrak{g}}_1$ and $\widetilde{\mathfrak{g}}_2$ are isomorphic to each other if and only if the respective cocycles ω_1 and ω_2 satisfy $\omega_1 - \omega_2 \in B^2(\mathfrak{g}; V)$. This isomorphism acts trivially on $\mathfrak{g}$ and V. Specifically if $\omega_1 - \omega_2 = d\varphi$ for some $\varphi \in \mathrm{Hom}(\mathfrak{g}, V)$ then the isomorphism is given by $(x, a) \mapsto (x, a + \varphi(x))$.

Proposition 2.2. $H^2(\mathfrak{g}; V)$ classifies all such abelian extensions $\widetilde{\mathfrak{g}}$ of $\mathfrak{g}$ by V modulo isomorphisms acting trivially on V and $\mathfrak{g}$. In essence it measures the obstuction to making an extension split, where distrinct cohomology classes represent inequivalent abelian extensions.

3. The Levi Decomposition

Without further ado, lettuce jump right into the heart of the talk...

Theorem 3.1 (Levi Decomposition)**.** Let $\mathfrak{g}$ be a Lie algebra over a field of characteristic zero. Then

$$\mathfrak{g} \cong \mathfrak{g}_{ss} \ltimes \operatorname{rad}(\mathfrak{g})$$

where $\mathfrak{g}_{ss} \subset \mathfrak{g}$ is the a semisimple subalgebra $\mathfrak{g}$.

Proof. $\operatorname{rad}(\mathfrak{g})$ is an ideal of $\mathfrak{g}$, so we make the quotient Lie algebra $\mathfrak{g}/\operatorname{rad}(\mathfrak{g}) =: \mathfrak{g}_{ss}$. Since we're working over the category of Lie algebras, there's no gauruntee that such an object splits nicely. But in the category of vector spaces, every subspace admits a linear complement. Hence *strictly* in the category of vector spaces we can form the right-split short exact sequence with linear section q where

$$\begin{tikzcd} 0 \arrow[r] & \operatorname{rad}(\mathfrak{g}) \arrow[r, hook, "i"] & \mathfrak{g} \arrow[r, two heads, "\pi"] & \mathfrak{g}_{ss} \arrow[r] \arrow[l, bend left, "q"] & 0 \end{tikzcd}$$

such that $\pi \circ q = \mathrm{Id}_{\mathfrak{g}_{ss}}$. Then we can represent any element in $\mathfrak{g}$ as the external direct sum $\operatorname{rad}(\mathfrak{g}) \times \mathfrak{g}_{ss}$ via $(a, x) \mapsto a + q(x)$, from which we can re-express the Lie bracket $[\cdot, \cdot]_{\mathfrak{g}}$. For $a, b \in \operatorname{rad}(\mathfrak{g})$ and $x, y \in \mathfrak{g}_{ss}$ we have by bilinearity,

$$[(a, x), (b, y)]_{\mathfrak{g}} = [a, b]_{\mathfrak{g}} + [q(x), b]_{\mathfrak{g}} + [a, q(y)]_{\mathfrak{g}} + [q(x), q(y)]_{\mathfrak{g}}.$$

Since $\operatorname{rad}(\mathfrak{g})$ is an ideal, the first three terms all land in $\operatorname{rad}(\mathfrak{g})$. The final term $[q(x), q(y)]_{\mathfrak{g}}$ is where q may fail to be a Lie algebra homomorphism: we meaure this failure with a bilinear map $\omega \colon \mathfrak{g}_{ss} \times \mathfrak{g}_{ss} \to \operatorname{rad}(\mathfrak{g}) = \ker \pi$ given by

$$\omega(x, y) = [q(x), q(y)]_{\mathfrak{g}} - q([x, y]_{\mathfrak{g}_{ss}}).$$

To see that $\operatorname{im}(\omega) = \ker(\pi)$ note that

$$(\pi \circ \omega)(x, y) = [\pi(q(x)), \pi(q(y))]_{\mathfrak{g}} - \pi(q([x, y]_{\mathfrak{g}_{ss}})) = [x, y] - [x, y] = 0.$$

To trap this error term, we induct on the dimension of $\operatorname{rad}(\mathfrak{g})$. The base case is trivial; if $\dim(\operatorname{rad}(\mathfrak{g})) = 0$, then $\mathfrak{g}$ is semisimple and the Levi decomposition holds on the nose. Assume now that $\dim(\operatorname{rad}(\mathfrak{g})) > 0$, then since $\operatorname{rad}(\mathfrak{g})$ is solvable there exists an $(n+1) \in \mathbb{N}$ satisfying $D^{n+1} = [D^n, D^n] = 0$ making D^n an abelian ideal of $\mathfrak{g}$.

Lemma 3.1.1. $\operatorname{rad}(\mathfrak{g}/D^n) = \operatorname{rad}(\mathfrak{g})/D^n$.

Proof. Letting $p \colon \mathfrak{g} \to \mathfrak{g}/D^n$ be the canonical projection, since $\operatorname{rad}(\mathfrak{g})$ is solvable, the image $\operatorname{rad}(\mathfrak{g})/D^n$ is a solvable ideal in the quotient. Supposing there's a larger solvable ideal $I \subset \mathfrak{g}/D^n$, the preimage $p^{-1}(I)$ would be an ideal in $\mathfrak{g}$ fitting into the short exact sequence

$$0 \longrightarrow D^n \longrightarrow \pi^{-1}(I) \longrightarrow I \longrightarrow 0.$$

However an extension of a solvable Lie algebra by a solvable Lie algebra is itself solvable, whence making $\pi^{-1}(I)$ a solvable ideal in $\mathfrak{g}$ strictly containing $\operatorname{rad}(\mathfrak{g})$, a contradiction to maximality. $\square$

By application of the Third Isomorphism Theorem,

$$(\mathfrak{g}/D^n)_{ss} = \frac{\mathfrak{g}/D^n}{\operatorname{rad}(\mathfrak{g})/D^n} \cong \mathfrak{g}/\operatorname{rad}(\mathfrak{g}) = \mathfrak{g}_{ss}.$$

Since $\dim(D^n) > 1$ we know that $\dim(\operatorname{rad}(\mathfrak{g})/D^n) < \dim(\operatorname{rad}(\mathfrak{g}))$ so by application of the inductive hypothesis, $\mathfrak{g}/D^n$ admits a valid Lie algebra section $\widetilde{q} \colon \mathfrak{g}_{ss} \to \mathfrak{g}/D^n$ yielding the diagram:

$$\begin{tikzcd} 0 \arrow[r] & \operatorname{rad}(\mathfrak{g}) \arrow[r, hook, "i"] \arrow[d, two heads] & \mathfrak{g} \arrow[r, two heads, "\pi"] \arrow[d, two heads, "p"] & \mathfrak{g}_{ss} \arrow[r] \arrow[d, "="] \arrow[l, bend right, "q"'] & 0 \\ 0 \arrow[r] & \operatorname{rad}(\mathfrak{g})/D^n \arrow[r, hook] & \mathfrak{g}/D^n \arrow[r, two heads] & \mathfrak{g}_{ss} \arrow[r] \arrow[l, bend left, "\widetilde{q}"] & 0 \end{tikzcd}$$

Since the top splitting is still just in the category of vector spaces, we can refine q as follows: choose a vector space basis for $\mathfrak{g}_{ss}$, the image of each basis element under $\widetilde{q}$ is an equivalence class in $\mathfrak{g}/D^n$ which, by surjectivity of p, has a representative in $\mathfrak{g}$ which we declare to be the image under q. Extend by linearity to the whole space, and $p \circ q = \widetilde{q}$, making q a lift of $\widetilde{q}$.

This new section $\widetilde{q}$ is of course a Lie algebra homomorphism; the bottom sequence splits in the category of Lie algebras. Hence the projection of our obstruction map must vanish: $p(\omega) = 0$. Thus for this specific lift q, the image of ω is restricted entirely to $\ker(p) = D^n$.

Lemma 3.1.2. $\omega \in Z^2(\mathfrak{g}_{ss}; D^n)$.

Proof. D^n is of course an abelian ideal of $\mathfrak{g}$, $\mathfrak{g}_{ss} \curvearrowright D^n$ via adjoin action, endowing D^n we the structure of a $\mathfrak{g}_{ss}$-module defined by $x \cdot a = [q(x), a]_{\mathfrak{g}}$. By applying the Jacobi identity in $\mathfrak{g}$ to lifted elements, we get that

$$0 = [q(x), [q(y), q(z)]_{\mathfrak{g}}]_{\mathfrak{g}} + [q(y), [q(z), q(x)]_{\mathfrak{g}}]_{\mathfrak{g}} + [q(z), [q(x), q(y)]_{\mathfrak{g}}]_{\mathfrak{g}}.$$

The first term can be replaced by the obstruction term definition $[q(x), q(y)]_{\mathfrak{g}} = q([x, y]_{g_{ss}}) + \omega(x, y)$ such that

$$\begin{aligned}
[q(x), [q(y), q(z)]_{\mathfrak{g}}]_{\mathfrak{g}} &= [q(x), q([y, z]_{g_{ss}}) + \omega(y, z)]_{\mathfrak{g}} \\
&= [q(x), q([y, z]_{g_{ss}})]_{\mathfrak{g}} + [q(x), \omega(y, z)]_{\mathfrak{g}} \\
&= \underbrace{q\big([x, [y, z]_{\mathfrak{g}_{ss}}]_{\mathfrak{g}_{ss}}\big)}_{=0} + \omega(x, [y, z]_{\mathfrak{g}_{ss}}) + x \cdot \omega(y, z)
\end{aligned}$$

where the first term vanishes from the Jacobi identity in $\mathfrak{g}_{ss}$. Applying this reformulation to each term in the previous equation, we know that

$$0 = \sum_{\text{cyclic}} \left(\omega(x, [y, z]_{\mathfrak{g}_{ss}}) - x \cdot \omega(y, z)\right) = -d_2\omega(x \wedge y \wedge z)$$

whence $\omega \in Z^2(\mathfrak{g}_{ss}; D^n)$. $\square$

You can probably see the next step coming from a mile away! Since $\mathfrak{g}_{ss}$ is semisimple and D^n is a finite-dimensional $\mathfrak{g}_{ss}$-module, by application of Whitehead's Theorem, $H^2(\mathfrak{g}_{ss}; D^n) = 0$. Hence there exists an $\eta \in Z^1(\mathfrak{g}_{ss}; D^n)$ such that $\omega = d_1\eta$. Explicitely,

$$\omega(x, y) = d_1\eta(x, y) = x \cdot \eta(y) - y \cdot \eta(x) - \eta([x, y]_{\mathfrak{g}_{ss}}).$$

We now define our *corrected* linear section $q' \colon \mathfrak{g}_{ss} \to \mathfrak{g}$ via the mapping $q'(x) = q(x) - \eta(x)$. Evaluating the commutator of this new guy yields

$$\begin{aligned}
[q'(x), q'(y)]_{\mathfrak{g}} &= [q(x) - \eta(x), q(y) - \eta(y)]_{\mathfrak{g}} \\
&= [q(x), q(y)]_{\mathfrak{g}} - [q(x), \eta(y)]_{\mathfrak{g}} - [\eta(x), q(y)]_{\mathfrak{g}} + [\eta(x), \eta(y)]_{\mathfrak{g}}.
\end{aligned}$$

Of course D^n is abelian, so $[\eta(x), \eta(y)]_{\mathfrak{g}} = 0$, whence we have

$$\begin{aligned}
[q'(x), q'(y)]_{\mathfrak{g}} &= q([x, y]_{\mathfrak{g}_{ss}}) + \omega(x, y) - x \cdot \eta(y) + y \cdot \eta(x) \\
&= q([x, y]_{\mathfrak{g}_{ss}}) - \eta([x, y]_{\mathfrak{g}_{ss}}) \\
&= q'([x, y]_{\mathfrak{g}_{ss}}).
\end{aligned}$$

Alas, q' is a true Lie algebra homomorphism so its image $q'(\mathfrak{g}_{ss})$ is a valid Levi subalgebra, making $\mathfrak{g} \cong q'(\mathfrak{g}_{ss}) \ltimes \mathrm{rad}(\mathfrak{g})$ and completing the proof. $\square$

References

[Eti24] Pavel Etingof, *Lie groups and Lie algebras*, 2024. ↑1

[Web13] Brian Weber, *Lecture 6: Lie Algebra Cohomology III*, 2013. Lecture notes for Math 651, University of Pennsylvania. ↑